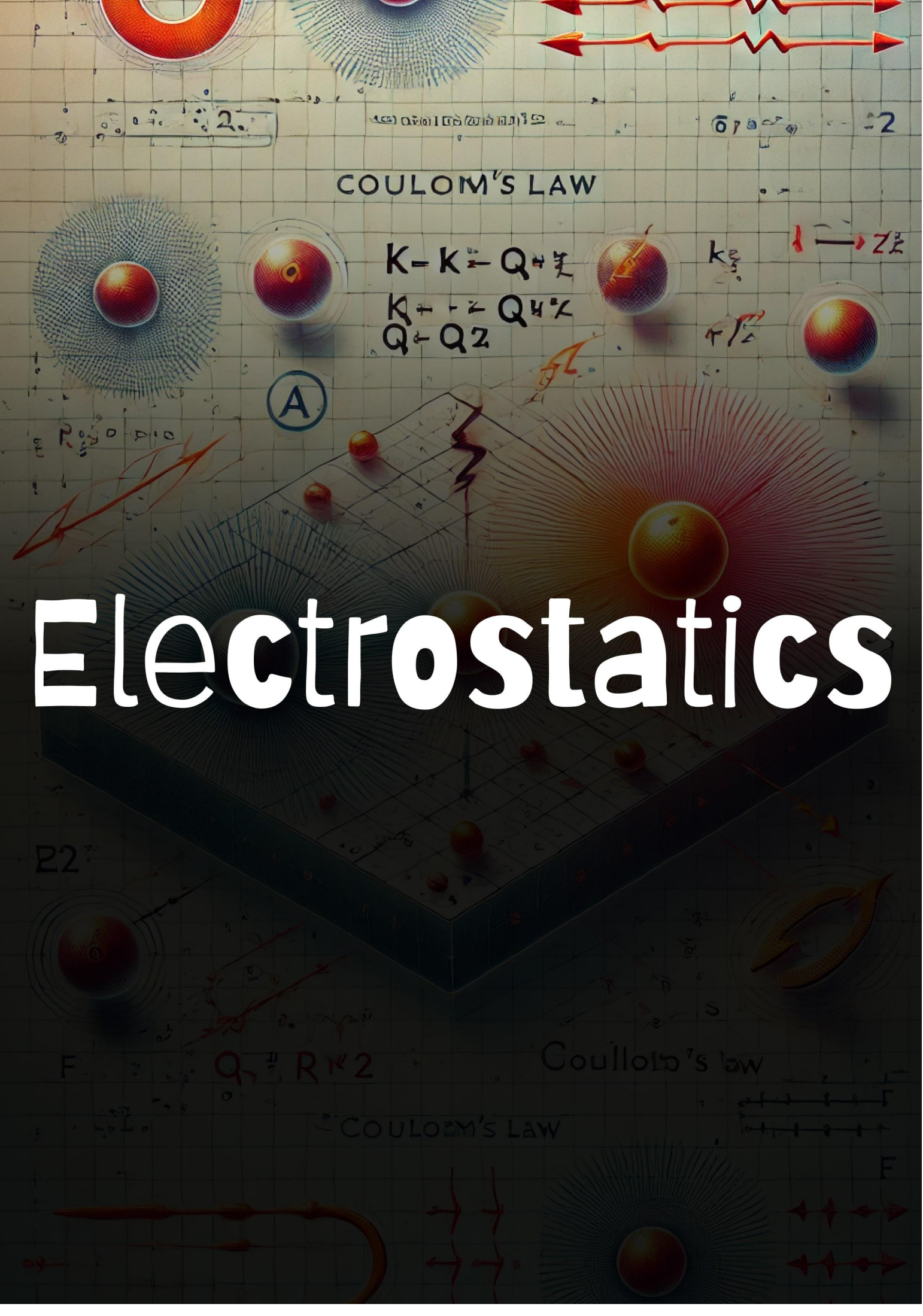

COULOM'S LAW
Electrostatics

Electrostatics

CHARGE

Fundamental property of matter (just like mass)
Unit ⟶ Coulomb's

NOTE: Charge is always quantised
(integer multiple of a certain no.)
Charge = Integer × (1.6×10^{-19}) Coulomb's
* like charges repel & unlike attract.

COULOMB'S LAW

The force of interaction b/w any two charges is directly proportional to the product of charges & inversely proportional to the square of distance b/w them.

Mathematically,

$$F = \frac{K q_1 q_2}{r^2}$$

Here q_1 and q_2 are the two charges & r is the separation b/w them and K is the proportionality constant.

$$K = 9 \times 10^9 \ Nm^2$$

ELECTRIC FIELD (INTENSITY)

The electrostatic field at any point of space is defined as the force per unit charge.

Mathematically,

$$\vec{E} = \frac{\vec{F}}{q} \quad \text{or} \quad q\vec{E} = \vec{F}$$

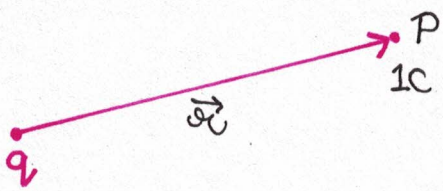
Where $\vec{F}$ is the force experienced by the charge q.

Unit ⟶ NC^{-1}

FIELD DUE TO A POINT CHARGE

$$\vec{E} = \frac{Kq}{r^2}\hat{r}$$

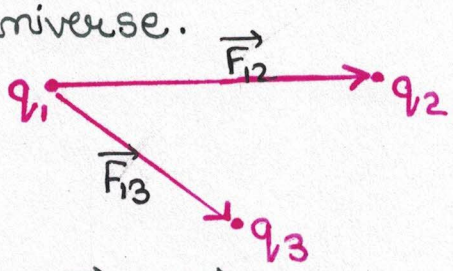
$$\vec{E} = \frac{Kq}{r^3}\vec{r}$$



421

SUPER - POSITION PRINCIPLE

The force of interaction b/w any two point charges is independent of the presence or absence of any other charge in the universe.

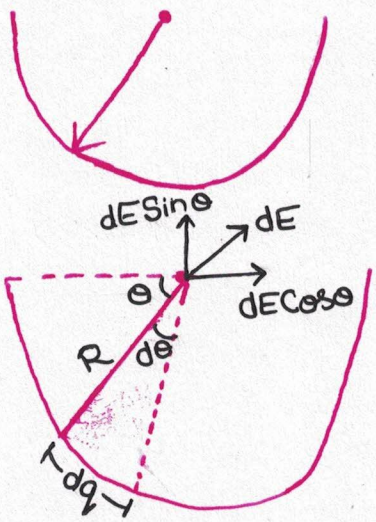


$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13}$$

FINDING FIELD BY INTEGRATION

Since field is a vector, the net field should be found by integrating the components of the elemental field separately.

Que. A thin semiconductor wire has linear charge density λ. What is the total electric field at point O due to this?



$$dq = \lambda R\, d\theta$$

$$dE = \frac{K dq}{R^2} = \frac{K\lambda d\theta}{R}$$

$$dE_y = dE \sin\theta$$

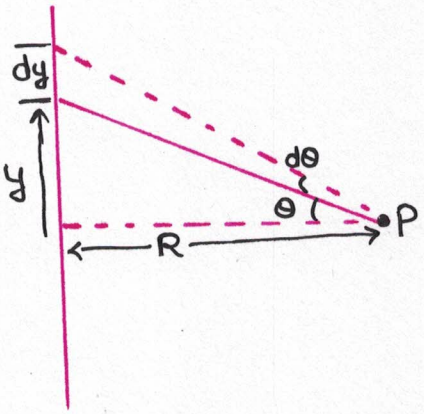
$$\int_0^{E_y} dE_y = \frac{K\lambda}{R} \int_0^{\pi} \sin\theta . d\theta$$

$$E_y = \frac{2K\lambda}{R} \quad , \quad E_x = 0$$

$$E = \frac{2K\lambda}{R} \qquad \text{(upwards)}$$

Que.) Find the field due to an infinite wire of charge having charge density λ at a distance R from it.

$$dE = \frac{K\lambda dy}{R^2 \sec^2\theta}$$



$$dE_x = dE\cos\theta$$

$$= \frac{K\lambda dy}{R^2\sec^2\theta}\cdot\cos\theta$$

$$y = R\tan\theta$$

$$dy = R\sec^2\theta.d\theta$$

$$dE_x = \frac{K\lambda R\sec^2\theta.d\theta.\cos\theta}{R^2\sec^2\theta}$$

$$= \frac{K\lambda}{R}\cos\theta.d\theta$$

$K = \frac{1}{4\pi\epsilon_0}$; ϵ_0 = permittivity of free space

$$\int dE_x = \frac{+K\lambda}{R}\int_{-\pi/2}^{\pi/2}\cos\theta.d\theta$$

$$E_x = \frac{K\lambda}{R}[\sin\theta]_{-\pi/2}^{\pi/2}$$

$$E_x = \frac{2K\lambda}{R}, \quad E_y = 0$$

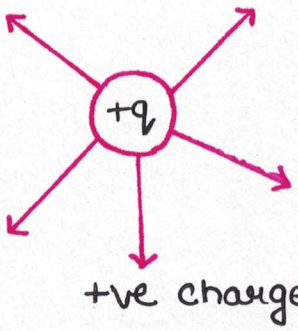
$$\therefore E = \frac{2K\lambda}{R}$$

ELECTRIC FIELD LINES

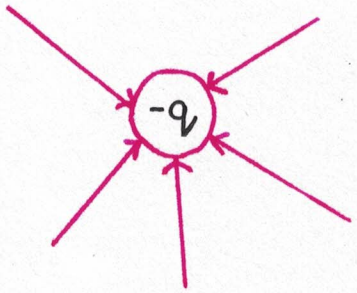
Field lines are a visual representation of the electric field. Their density is proportional to the magnitude of the field and their direction (tangent) gives the direction of the field.

NOTE: Electric field lines in general do not represent the path followed by the particle.

Examples:

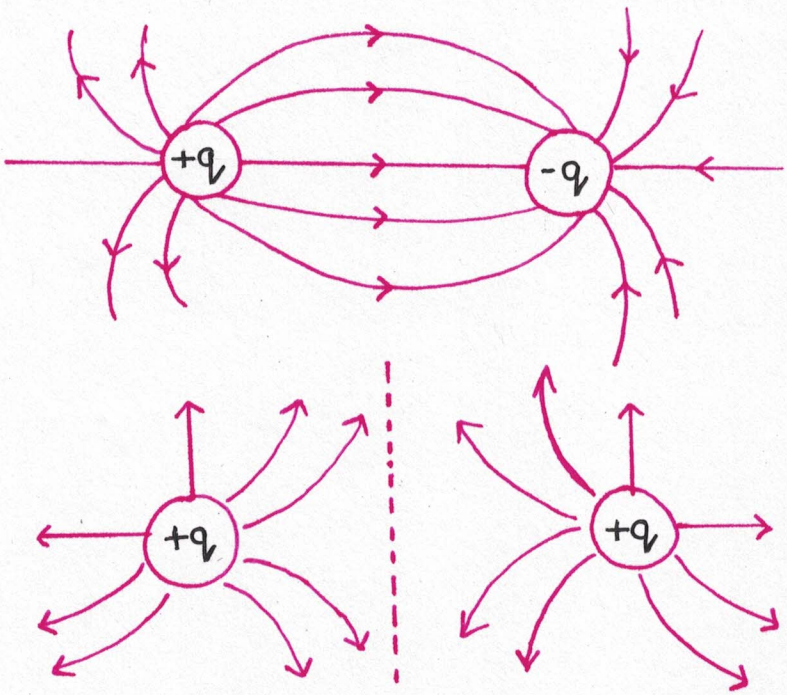


+ve charge



-ve charge

ELECTRIC DIPOLE FIELD PATTERN



In middle, $E = 0$

at other points, E can be calculated

SOME PROPERTIES OF FIELD LINES

1.) Field lines eminate from a +ve charge (or infinity) & terminate on a -ve charge (or infinity).

2.) Field lines never intersect (or touch) each other because the force direction has to be unique.

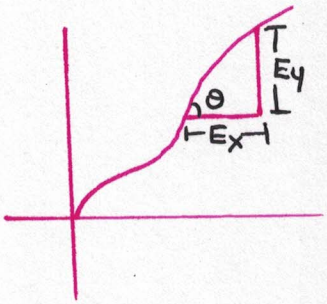
3.) Since the electrostatic field is conservative, the field lines for an 'electrostatic field' never form a closed loop.

424

Que:) The electric field in a certain region of space is given by $(\vec{E} = x\hat{i} + 2y\hat{j})$

Find the eqn. of field line passing through the point (1,2).

$$\tan\theta = \frac{dy}{dx} = \frac{E_y}{E_x} = \frac{2y}{x}$$



$$\Rightarrow \frac{dy}{dx} = \frac{dx}{x}$$

$$\frac{1}{2}\int y^{-1}dy = \int x^{-1}dx$$

$$\frac{1}{2}\ln(y) = \ln(x) + C$$

$$\ln(y) = 2\ln(x) + 2C$$

$$\ln\left(\frac{y}{x^2}\right) = C'$$

$$\frac{y}{x^2} = e^{C'} = K$$

$$e^C = 2$$

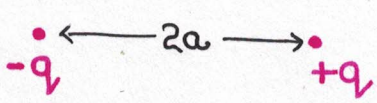
$$K = 2 = e^C$$

equation $\Rightarrow$ $y = 2x^2$

DIPOLE MOMENT

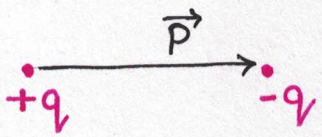
Two equal and opposite charges separated by a small distance from a dipole.

Eg:-



Dipole moment is a vector directed from -ve charge to the +ve charge and its magnitude is the product of separation b/w the charges & magnitude of the charges.

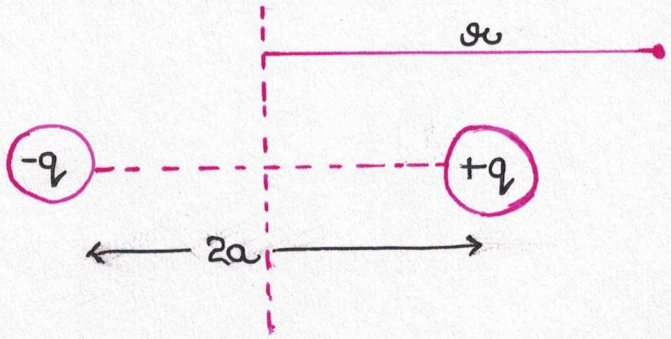
Mathematically, $\vec{p} = 2qa$



NOTE: Point Dipole

It is a mathematical idealisation where the separation b/w the charges is taken as arbitrary close to zero. At sufficiently large distances, as compared to the dimension of dipole, any dipole can be treated as a point dipole.

FIELD DUE TO THE DIPOLE AT AXIAL POSITION



$$\vec{E} = \frac{Kq}{(r-a)^2}\hat{i} + \frac{Kq}{(r+a)^2}(-\hat{i})$$

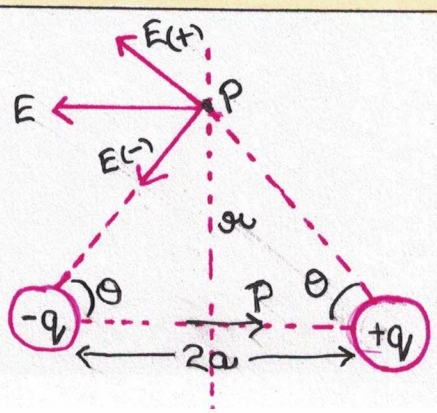
$$\text{or } \vec{E} = \frac{Kq(4ra)}{(r^2-a^2)^2}\hat{i}$$

$$\vec{p} = 2qa\hat{i}$$

$$\therefore \quad \boxed{\vec{E} = \frac{2Kr\vec{p}}{(r^2-a^2)^2}}$$

If $r >> a$ $\quad \vec{E} = \frac{2K\vec{p}}{r^3}$ (for point dipole)

FIELD DUE TO A DIPOLE ON EQUITORIAL POSITION



426

$$E = 2E_+ \cos\theta$$

$$E_+ = \frac{Kq}{(\sqrt{a^2+x^2})^2}$$

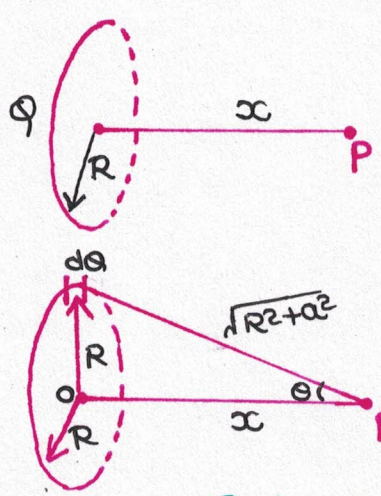
$$\cos\theta = \frac{a}{\sqrt{a^2+x^2}}$$

$$\therefore E = \frac{2Kqa}{(x^2+a^2)^{3/2}}(-\hat{i})$$

or $$\vec{E} = \frac{-K\vec{p}}{(x^2+a^2)^{3/2}}$$

If $x >> a$ $$\vec{E} = \frac{-K\vec{p}}{x^3}$$ (for point charge)

Que: A charge Q is uniformly distributed on a ring of radius R. Find field at point P on axis of ring at a distance x from the centre.



$$dE = \frac{K\,dQ}{(\sqrt{R^2+x^2})^2}$$

$$dE_x = dE\cos\theta = \frac{Kx}{(a^2+x^2)^{3/2}}dQ$$

$$E_x = \frac{Kx}{(R^2+x^2)^{3/2}}\cdot Q \quad , \quad E_y = 0$$

$$\therefore E = \frac{Kx}{(R^2+x^2)^{3/2}}Q$$

(Field is max at $x = \pm R/\sqrt{2}$)

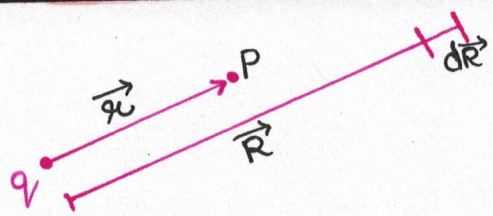
ELECTROSTATIC POTENTIAL

Negative of work done by electrostatic forces in bringing a unit charge from infinity to that point is called electrostatic potential at that point.

(Same as work done in taking that charge from that point to infinity.)

427

POTENTIAL DUE TO A POINT CHARGE



$$\vec{E} = \frac{Kq}{R^2}\hat{R}$$

$$d\vec{R} = dR.\hat{R}$$

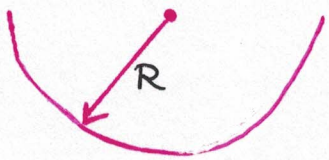
$$dV = dW = \vec{F}.d\vec{R} = \frac{Kq}{R^2}\hat{R}.d\vec{R}$$

$$V = \int dW = \int \frac{Kq}{R^2}.dR$$

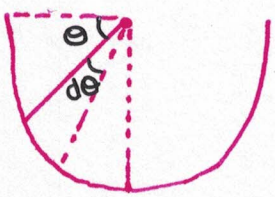
$$V = Kq\left[-\frac{1}{R}\right]_{r}^{\infty}$$

$$V = \frac{Kq}{r}$$

Que: Find potential at point O.



λ = Charge density



$$dV = \frac{K(\lambda R\, d\theta)}{R}$$

$$V = K\lambda\int_0^{\pi} d\theta$$

$$V = K\lambda\pi$$

Relation b/w field and Potential

$$dV = -\vec{E}.d\vec{r}$$

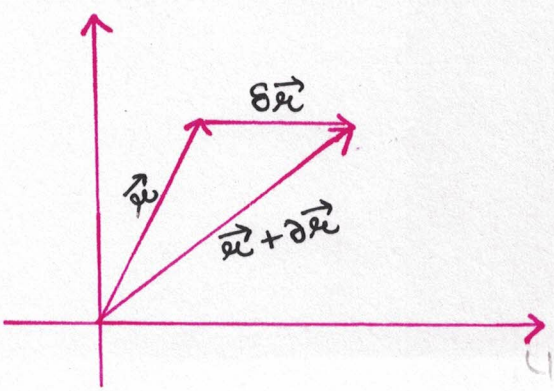
FINDING FIELD FROM POTENTIAL

$$\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k} \quad \text{(given)}$$

$$\partial\vec{r} = \partial x\hat{i} + \partial y\hat{j} + \partial z\hat{k}$$

$$\therefore \partial V = -\vec{E}.\partial\vec{r}$$

$$\partial V = -\left[E_x\partial x + E_y\partial y + E_z\partial z\right]$$



If we move along x-axis
then $\partial y = \partial z = 0$

$$\partial V = -E_x \partial x$$

or $$E_x = -\frac{\partial V}{\partial x}$$

Similarly, $$E_y = -\frac{\partial V}{\partial y}$$

& $$E_z = -\frac{\partial V}{\partial z}$$

Que. Given $V = (x^2y + y^2z)$ Volts
Find (a) Field function
(b) Force on a 9 μC charge kept at (1, 1, 1)
(c) Work done in moving this charge to the point (2, 2, 2) by electric field.

(a) $\vec{E} = [-2xy] + [-(x^2 + 2yz)] + [-(y^2)]$

(b) At (1, 1, 1), $\vec{E} = -2\hat{i} - 3\hat{j} - 1\hat{K}$

$\vec{F} = q\vec{E}$

$= (-18\hat{i} - 27\hat{j} - 9\hat{K})$ μN

(c) $\Delta W_E = -q\Delta V$

$= -(9\,\mu C)(V_B - V_A)$

$= -(9\,\mu C)[16 - 2]$

$= -126\,\mu C$ (volts)

$\Delta W = -126\,\mu J$

NOTE: When we move along a field line (i.e. in direction of arrow), potential decreases.

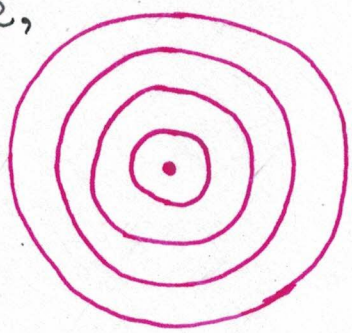
EQUIPOTENTIAL SURFACES

A surface which joins the points having some potential is known as an equipotential surface.

→ An equipotential surface is always ⊥ to the electric field lines.

For visualization of electric field, we can draw equipotential surfaces. If the surfaces are drawn near to each other it means large value of electric field.

For a point charge,



$$\text{Slope of equipotential surface} = -\frac{E_x}{E_y}$$

Que.) The field function in a region of space is given by

$$\vec{E} = x\hat{i} + 2y\hat{j}$$

Find eqⁿ. of equipotential surface passing through (1,1)

$$\frac{dy}{dx} = -\frac{x}{2y}$$

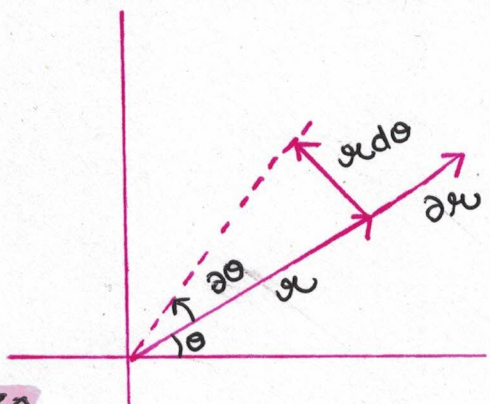
$$\int 2y\,dy = -\int x\,dx$$

$$\frac{x^2}{2} + y^2 = C$$

$$x^2 + 2y^2 = C$$

Through (1,1) $\Rightarrow$ $x^2 + 2y^2 = 3$

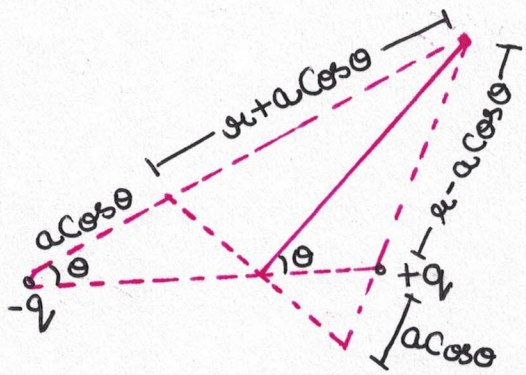
FINDING FIELD FROM POTENTIAL - POLAR FORM

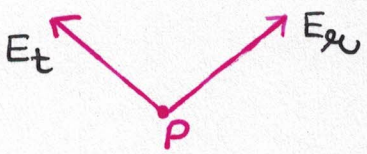


$$E_r = -\frac{\partial V}{\partial r}$$

$$E_\theta = -\frac{1}{r}\frac{\partial V}{\partial \theta}$$

Que.) Find the potential due to a dipole at an arbitrary location (r, θ). Using this, find the field at arbitrary location & verify your result for axial & equitorial positions. Assume small dipole $(a << R)$





$$V_p = \frac{Kq}{(r - a\cos\theta)} - \frac{Kq}{(r + a\cos\theta)}$$

$$V = \frac{2Kq\, a\cos\theta}{r^2 - a^2\cos^2\theta}$$

Since $a << R$ $\therefore V = \frac{2Kq\, a\cos\theta}{r^2}$

$$V = \frac{Kp\cos\theta}{r^2}$$

$$E_r = -\frac{\partial V}{\partial r} = \frac{2Kp\cos\theta}{r^3}$$

$$E_\theta = -\frac{1}{r}\frac{\partial V}{\partial \theta} = \frac{Kp\sin\theta}{r^3}$$

$$E = \sqrt{E_r^2 + E_\theta^2} = \frac{Kp}{r^3}\sqrt{1 + 3\cos^2\theta}$$

$$E = \frac{Kp}{r^3}\sqrt{1 + 3\cos^2\theta}$$

At axial, $\theta = 0^\circ$

$$E_a = \frac{2Kp}{r^3}$$

At equitorial, $\theta = 90^\circ$

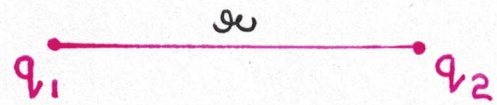
$$E_e = \frac{Kp}{r^3}$$

431

ELECTROSTATIC POTENTIAL ENERGY

For a system of particles, -ve of work done by electric field in bringing all the particles from ∞ to the given configuration is called E.P.E. of the configuration.

Que.) Find potential energy of the system of 2 particles shown in fig.



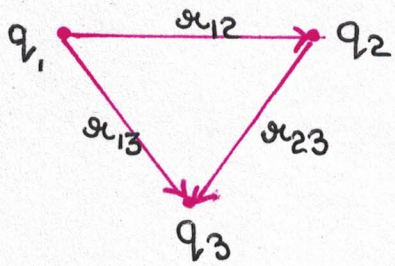
$$P.E. = \frac{Kq_1 \cdot q_2}{r} = -\Delta W$$

let both q_1 & q_2 be at ∞

q_1 can be brought freely

when q_2 is brought, work is done as q_1 is already present.

For a three particle system,



$$P.E. = \frac{Kq_1q_2}{r_{12}} + \frac{Kq_2q_3}{r_{23}} + \frac{Kq_3q_1}{r_{13}}$$

For a system of n-particles

$$P.E. = K \sum_{0<i,j\leq n} \frac{q_i \cdot q_j}{r_{ij}}$$

Consider each pair only once.

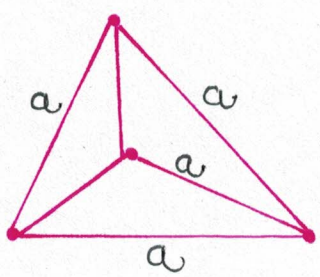
Que.) Find PE of the system

$q, q, a-1$

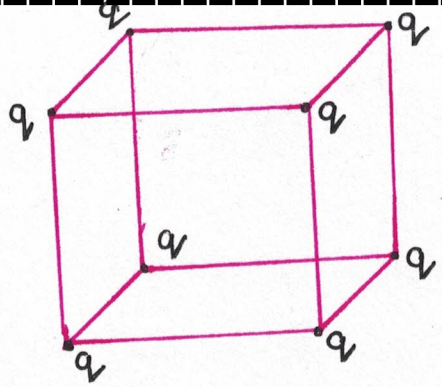
$q, -q, a-4$

$-q, -q, a-1$

$$PE = -\frac{2Kq^2}{a}$$



Que.



$q, q, a-9$

$q, q, a\sqrt{2}-9$

$q, q, \sqrt{3}a-3$

$q, -q, a-3$

$q, -q, \sqrt{2}a-3$

$q, -q, \sqrt{3}a-1$

$$PE = \frac{6Kq^2}{a} + \frac{6Kq^2}{\sqrt{2}a} + \frac{2Kq^2}{\sqrt{3}a}$$

ELECTRIC FLUX

Area Vector: For any small surface, having surface area dS, the area vector is defined as a vector whose magnitude is equal to dS & direction is ⊥ to dS.

Let $\vec{E}$ be the electric field associated with the small area vector $d\vec{S}$, then the small amount of electric flux associated with this patch is given by

$$d\phi = \vec{E} \cdot d\vec{S}$$

For large surface, flux (total) can be found by integrating the flux of its patches.

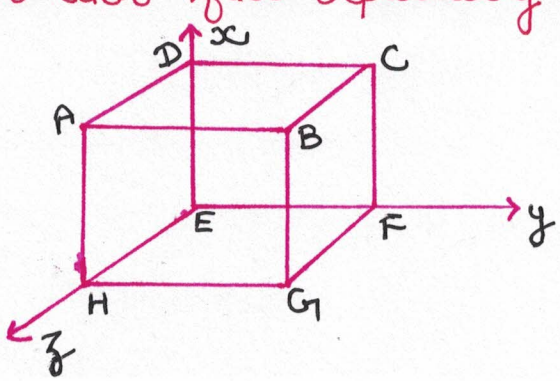
NOTE: For open surfaces, the direction of normal can be arbitrarily chosen, However, for closed surfaces, normal always means outwards normal.

Que.) Electric field in a region is given by

$$\vec{E} = 3\hat{i} + 2\hat{j} + \hat{k}$$

→ Find flux associated with the cube ABCDEFGH whose edges are parallel to the coordinate axes.

→ Find flux for each face separately given edge size = 2m.



433

ABCD	$\vec{S} = 4\hat{j}$	$\Phi = 8\ Nm^2C^{-1}$
BCFG	$\vec{S} = 4\hat{i}$	$\Phi = 12\ Nm^2C^{-1}$
EFGH	$\vec{S} = -4\hat{j}$	$\Phi = -8\ Nm^2C^{-1}$
ADEH	$\vec{S} = -4\hat{i}$	$\Phi = -12\ Nm^2C^{-1}$
ABGH	$\vec{S} = 4\hat{k}$	$\Phi = 4\ Nm^2C^{-1}$
DCFE	$\vec{S} = -4\hat{k}$	$\Phi = -4\ Nm^2C^{-1}$

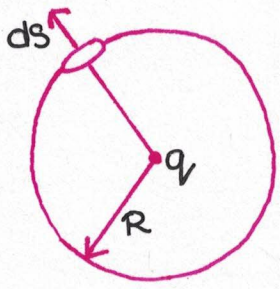
Total flux (Φ_T) = 0



NOTE: Electric flux is proportional to the no. of field lines associated with an area.

Zero Flux: No. of field lines entering is equal to no. of field lines coming out.

Que.) Find flux associated with a sphere of radius R at its center.



$$E = \frac{Kq}{R^2}$$

$$d\Phi = E\,ds\cos\theta$$

$$\theta = 0^\circ$$

$$d\Phi = \frac{Kq}{R^2}\,ds$$

$$\Phi = \frac{Kq}{R^2}\int ds = \frac{Kq}{R^2}\,4\pi R^2$$

$$\Phi = \frac{1}{4\pi\epsilon_0}\cdot q\cdot 4\pi$$

$$\Phi = \frac{q}{\epsilon_0}$$

GAUSS LAW

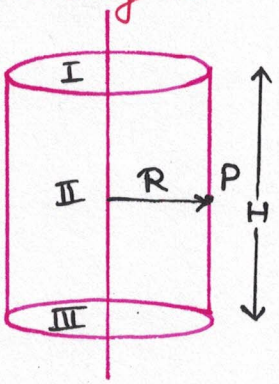
Net flux associated with any closed surface is equal to the charge enclosed, divided by ϵ_0.

Mathematically,

$$\Phi = \oint \vec{E}\cdot d\vec{S} = \frac{q_{enclosed}}{\epsilon_0}$$

434

Que.) Using Gauss law, find field due to an infinite wire having charge density λ.



Cylindrical gaussian surface

$\Phi_I = 0 \quad , \quad \Phi_{III} = 0$

$\Phi_{II} = (E)(2\pi RH)$

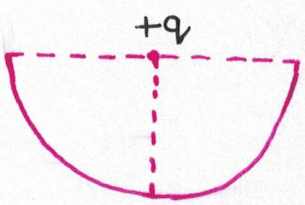
$\Phi_{net} = \frac{q_{enclosed}}{\epsilon_0}$

$2\pi RHE = \frac{\lambda H}{\epsilon_0}$

$E = \frac{\lambda}{2\pi\epsilon_0 R} = \frac{2K\lambda}{R}$

$E = \frac{2K\lambda}{R}$

Que.) Find flux associated



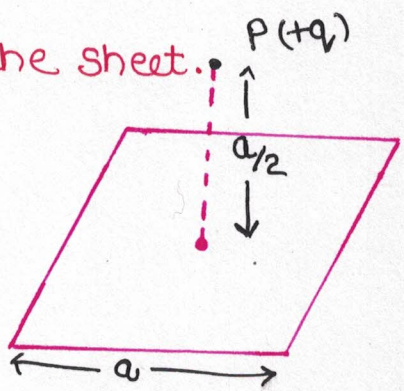
let flux associated $= \phi$

If we place another hemisphere, we get closed sphere.

$\therefore 2\phi = \frac{q}{\epsilon_0}$

$\phi = \frac{q}{2\epsilon_0}$

Que.) What will be the flux through the sheet.



Place 5 more sheets

$6\Phi = \frac{q}{\epsilon_0}$

$\Phi = \frac{q}{6\epsilon_0}$

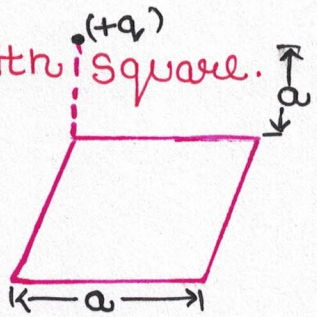
★ If the charge is placed just above the sheet, then $\phi = q/2\epsilon_0$

435

★ If the charge is placed on the sheet, then $\phi = 0$

($\because \vec{E}.d\vec{S} = 0$ as $\vec{E} \perp d\vec{S}$)

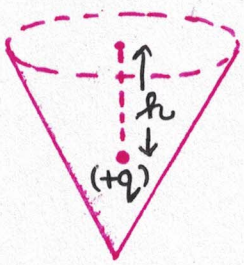
Que. Find flux associated with square.



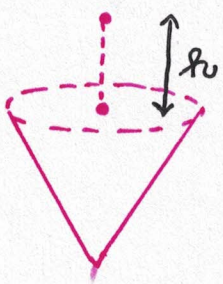
$$\phi = \frac{q}{24\epsilon_0}$$

Que. Find flux passing through Cone.

$$\phi = \frac{2q}{3\epsilon_0}$$

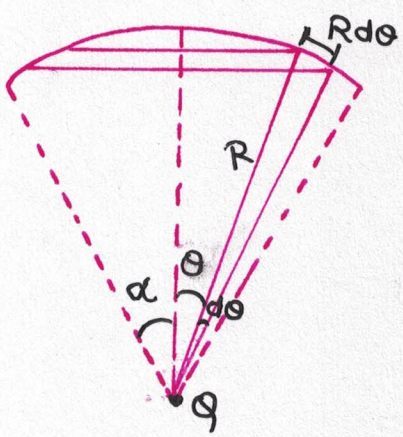


Que. What will be the flux now?



$$\phi = \frac{q}{3\epsilon_0}$$

Que. Find flux associated with the cap of sphere having semi-vertex angle α and charge Q placed at center of curvature.



$$d\phi = \left(\frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{R^2}\right)$$

$$(2\pi R)(\sin\theta . R d\theta)$$

$$\phi = \frac{Q}{2\epsilon_0}\int_0^{\alpha} \sin\theta . d\theta$$

$$\phi = \frac{Q}{2\epsilon_0}[-\cos\theta]_0^{\alpha}$$

$$\phi = \frac{Q}{2\epsilon_0}(1 - \cos\alpha)$$

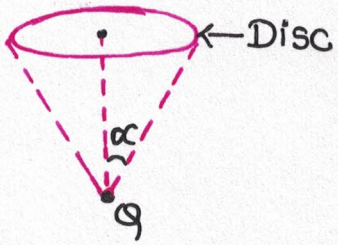
★(Flux $\propto$ No. of field lines)

436

i.e. If no. of field lines is same, then the flux is same.

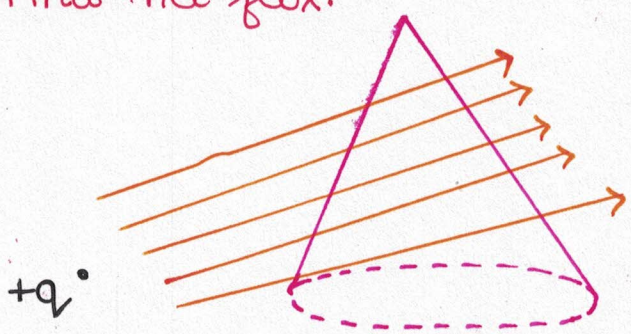
Que:) Find flux associated with the disc, subtending semi-vertex angle α at a charge q.

$$\phi = \frac{q}{2\epsilon_0}(1 - \cos\alpha)$$



★ Each entering field line is counted as -1 and each leaving field line is counted as +1.

Que:) Find net flux.



Since each field line which is entering the cone is leaving out, $\therefore \phi = 0$

★ If a closed surface does not enclose a charge, its flux contribution is zero.

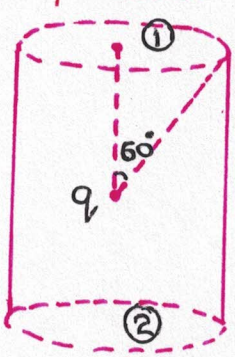
Que:) Find flux associated with C.S.A. of cylinder.

For surface ①,

$$\phi_1 = \frac{q}{4\epsilon_0}$$

Similarly,

$$\phi_2 = \frac{q}{4\epsilon_0}$$

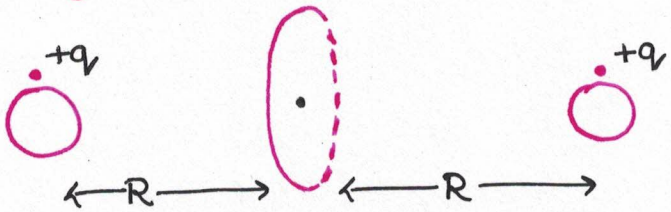


for whole cylinder, $\phi_n = \frac{q}{\epsilon_0}$

$$\therefore \phi_{CSA} = \frac{q}{2\epsilon_0} = \frac{q}{\epsilon_0} - \frac{q}{4\epsilon_0} - \frac{q}{4\epsilon_0}$$

437

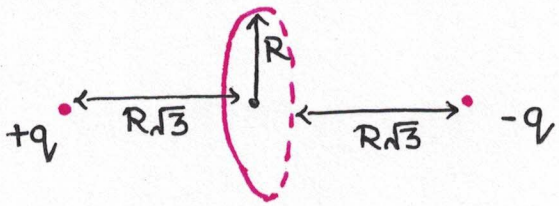
Que.) Disc of radius R.



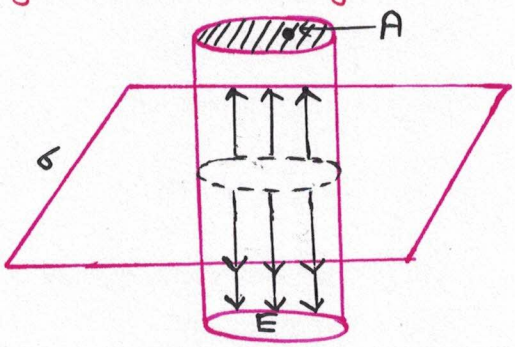
If we assume area normal to be on right then field due to (1) will make acute angle with area normal & (2) will form obtuse angle with same magnitude.

$\therefore \phi = 0$

Que.) Find flux associated with disc.



Que.) Derive an expression for field due to an infinite sheet of charge having areal charge density σ.



Consider a gaussian surface in form of cylinder.

$$2EA = \frac{\sigma A}{\epsilon_0} \quad \Rightarrow \quad E = \frac{\sigma}{2\epsilon_0}$$

GAUSS LAW FOR GRAVITATION

$$\oint \vec{g} \cdot d\vec{S} = -4\pi G m_{enclosed}$$

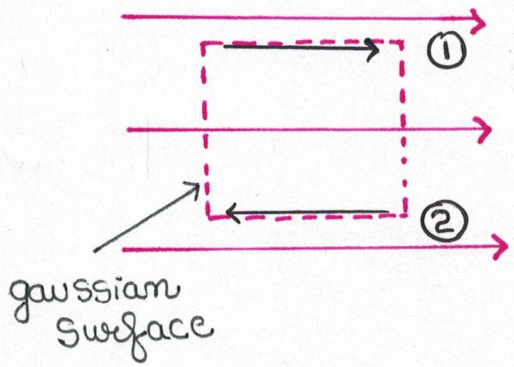
$$\because -\frac{GM}{r^2} = \frac{Kq}{r^2}$$ (-ve bcoz force is always negative)

$$-G = \frac{1}{4\pi G}$$

$$\frac{1}{\epsilon_0} = -4\pi G$$

438

Que.) Can the shown field lines represent an electrostatic field?



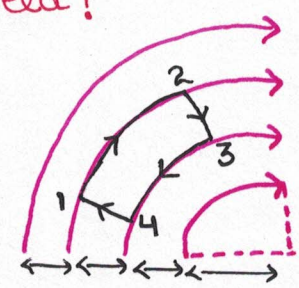
$W_1 = x$

$W_2 = -y$ $\quad (y > x)$

No!

$\therefore W_{net} = -ve$

Que.) Can the shown field lines represents an electrostatic field?



No!

Consider a path

From 1 to 2, $W = (+ve) \times$ (q × distance)

From 2 to 3 & from 4 to 1, $W = 0$

From 3 to 4, $W = -y$

$(x > y)$ $\quad \therefore W_{net} = +ve$

PROOF OF SHELL THEOREMS USING GAUSS LAW

SHELL THEOREM 1: For an outside point, a uniformly charged thin shell behaves as if all the charge were concentrated at its centre.

SHELL THEOREM 2: For an inside point, a uniformly charged thin spherical shell behaves as if there is no charge.

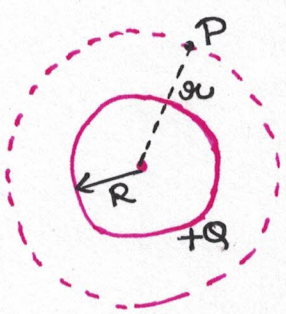
Proof of Shell Theorem 1:

$r > R$

Let field = E

$$E \times 4\pi r^2 = \frac{Q}{\epsilon_0}$$

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r^2}$$



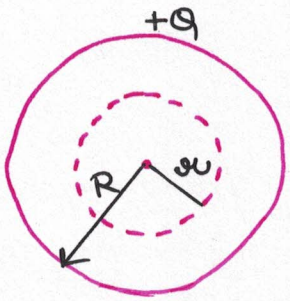
(Same as field of point charge)

Q placed at centre of the shell.

439

Proof of Shell Theorem 2:

$r < R$



$E \times 4\pi r^2 = 0$

$E = 0$

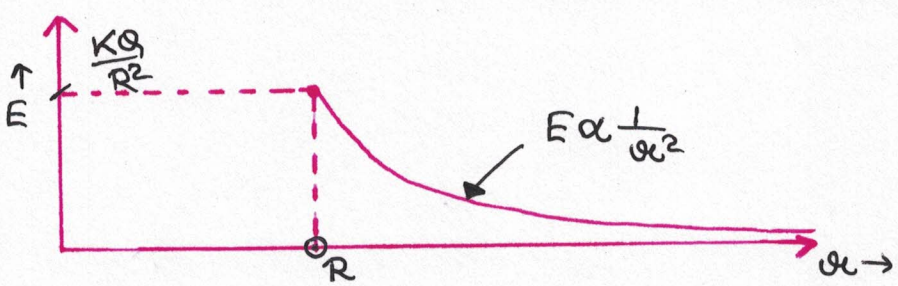
This is same as if there were no charge.

→ Contribution of shell is zero.
→ Field may not be zero.

★ $E = \text{zero} \Rightarrow \phi = \text{zero}$

$\phi = \text{zero} \not\Rightarrow E = \text{zero}$

VARIATION OF FIELD & POTENTIAL DUE TO A UNIFORMLY CHARGED THIN SHELL

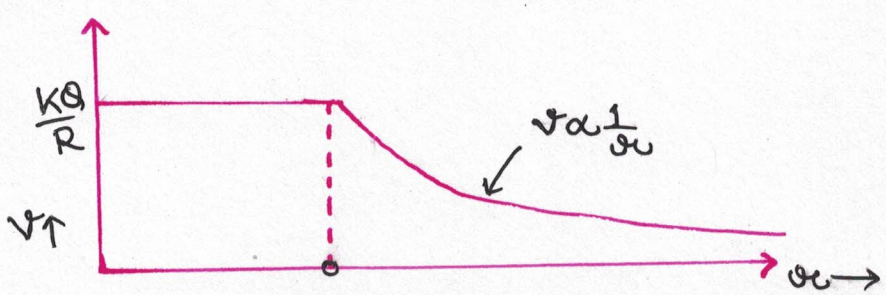


For an outside point, the shell behaves just like a point charge
∴ Formula for potential is also same as that for a point charge.

i.e. for $r > R$, $V = \frac{KQ}{r}$

& for $r \leq R$, $V = \frac{KQ}{R}$

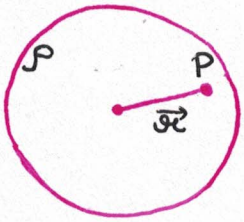
Potential is -ve of work done in bringing a unit charge from infinity. Inside the shell, field is zero,
∴ after crossing the surface, there is no work done in moving the charge inside. Thus, potential at inside point is same as that at the surface.



440

VARIATION OF FIELD & POTENTIAL FOR A UNIFORMLY CHARGED SOLID SPHERE

ρ = charge density



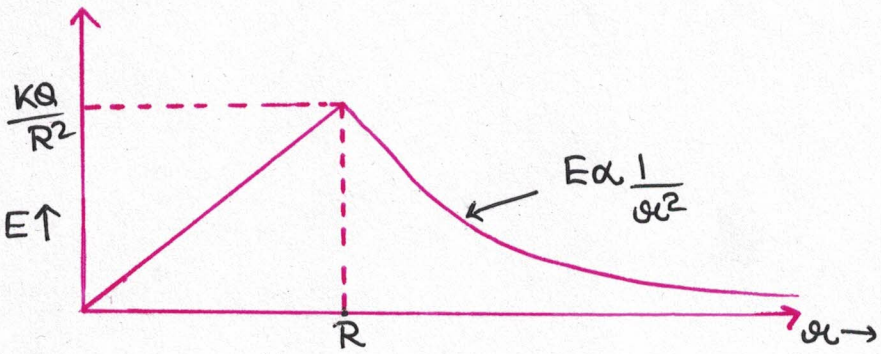
$r < R$, $E \times 4\pi r^2 = \rho \times \frac{4}{3}\pi r^3 \times \frac{1}{\epsilon_0}$

$$E = \frac{\rho r}{3\epsilon_0} = \frac{KQr}{R^3}$$

or $$\vec{E} = \frac{\rho}{3\epsilon_0}\vec{r}$$

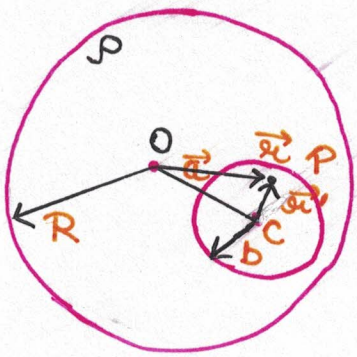
$r > R$, Sphere behaves as a point charge (∵ Shell Theorem)

$$E = \frac{KQ}{r^2}$$



Que: Prove that electric field in the cavity is constant & find its magnitude and direction.

$$\vec{E} = \frac{\rho}{3\epsilon_0}\vec{r}$$



Consider point P

$\vec{OC} = \vec{a}$, $\vec{OP} = \vec{r}$, $\vec{CP} = \vec{r'}$

Contribution of Sphere $= \frac{\rho}{3\epsilon_0}\vec{r}$

Contribution of cavity $= -\frac{\rho}{3\epsilon_0}\vec{r'}$

$$\vec{E}_{net} = \frac{\rho}{3\epsilon_0}(\vec{r} - \vec{r'})$$

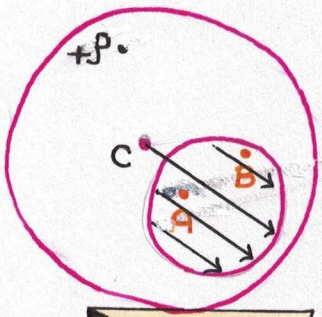
or $$\vec{E} = \frac{\rho}{3\epsilon_0}\vec{a}$$

{ Sphere is +vely charged with density ρ while cavity is neutral so density is $-\rho$

Que.) Which point is at higher potential : A or B ?

A is at higher potential

∵ when we move along field line, potential decreases.



POTENTIAL DUE TO A UNIFORMLY CHARGED SOLID SPHERE

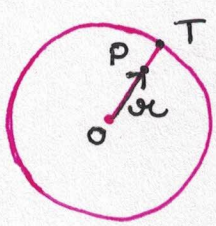
For outside point,

the entire sphere behaves like a point charge at its own center.

∴ Formula for potential is same as that for a point charge

i.e. for $x > R$, $V = \frac{KQ}{x}$

For inside point,



$$dV = -\vec{E}.d\vec{x}$$

$$\vec{E} = \frac{KQ}{R^3}\vec{x}, \quad d\vec{x} = dx.\hat{x}$$

$$\therefore dV = -\frac{KQ}{R^3} x\,dx$$

$$\int_{V_P}^{V_T} dV = -\frac{KQ}{R^3}\int_{x}^{R} x.dx$$

$$V_T - V_P = -\frac{KQ}{R^3}\left[\frac{R^2}{2} - \frac{x^2}{2}\right]$$

$$V_P = V_T + \frac{KQ}{R^3}\left[\frac{R^2}{2} - \frac{x^2}{2}\right]$$

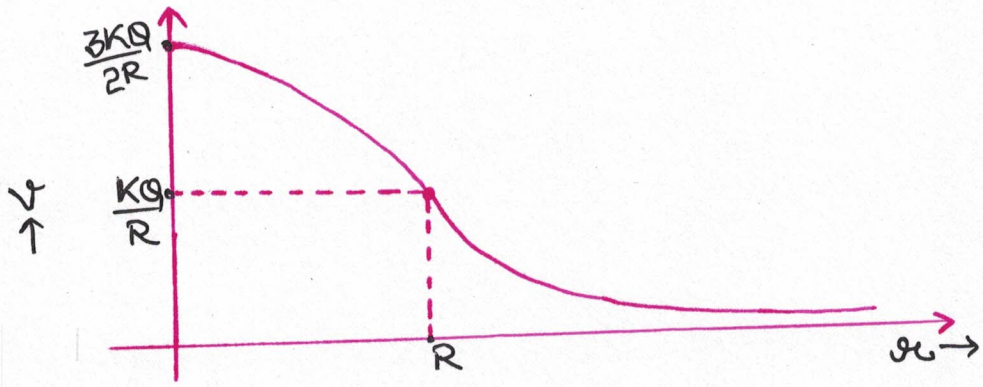
Now, $V_T = \frac{KQ}{R}$

$$\therefore V_P = \frac{KQ}{R} + \frac{KQ}{2R} - \frac{KQ\,x^2}{2R^3}$$

$$V_P = \frac{KQ}{2R}\left(3 - \frac{x^2}{R^2}\right)$$

$$V_P = \frac{KQ}{R}\left[1.5 - 0.5\frac{x^2}{R^2}\right]$$

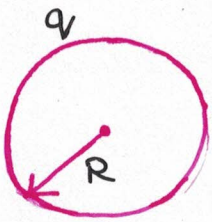
$$= \frac{KQ}{2R}\left[3 - \frac{x^2}{R^2}\right]$$



SELF - ENERGY

Negative of the work done by electrostatic forces in assembling an electrostatic system is called the self-energy of the electrostatic system.

Self energy of a uniformly charged shell charge q, has been brought from infinity & dq is to be brought.



$$V_{\text{at surface}} = \frac{Kq}{R}$$

$$dv = v.dq$$

$$\int_0^u dv = \int_0^{+q} \frac{Kq}{R}.dq$$

$$u = \frac{KQ^2}{2R}$$

Que.) Prove that for a uniformly charged solid sphere, the self energy is given by $u = \frac{3}{5}\frac{KQ^2}{R}$

443